SURYAGRID AI

Plant Technical Onboarding Dossier

Synthetic plant hierarchy, data readiness, sensor metadata and event library

Document Type	Plant Onboarding Dossier
Version	v1.0
Use	Sample PDF upload for LLM requirement extraction, planning and forecast design
Data Status	Synthetic demonstration - not real plant SCADA or commercial data
Primary Objective	Enable a model to infer business use case, horizon, variables, methods, metrics and data gaps

LLM extraction note: This document intentionally uses clear labels, tables and repeated field names so a document-reading model can extract business objectives, SCADA/MET/NWP variables, forecast horizons, metrics and RFP requirements with minimal ambiguity.

1. Plant Onboarding Summary

This dossier gives the onboarding agent the plant hierarchy, technical metadata, sensor deployment, data readiness, event history and model configuration needed to recommend a forecasting solution. All values are synthetic and provided for application demonstration.

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LLM-readable context: DOCUMENT_TYPE = PLANT_TECHNICAL_ONBOARDING_DOSSIER. HIERARCHY = GRID_PAVAGADA_220KV ->
SOL_PAVAGADA -> PAV_HUB_1..8 -> PAV_Sx_Byy. EXPECTED_AGENT_ACTION = extract hierarchy + data readiness + hub/block
events + forecast configuration.
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2. Plant Master Data

Field	Value
Plant name	Pavagada Solar Park - synthetic sample
Asset type	Solar PV
Design capacity	2050 MW AC / 2350 MWp DC
Grid node	Pavagada 220kV pooling station
Data history	10 years SCADA, 5 years met observations, 3 years NWP ensemble history
Business objective	Minimize deviation penalties, optimize grid dispatch and provide ramp warnings
Forecast hierarchy	Grid, plant, hub/segment and block level

3. Hub and Block Layout

Hub ID	Blocks	Capacity MW	Current status	Primary sensor package	Known event examples
PAV_HUB_1	PAV_S1_B01..B05	250	GOOD	POA, GHI, module temp, inverter status, SCADA actual/possible/schedule	normal historical operation
PAV_HUB_2	PAV_S2_B01..B05	250	STALE example on B01	POA, GHI, module temp, inverter status, SCADA actual/possible/schedule	scada_stale_signal on B01
PAV_HUB_3	PAV_S3_B01..B05	250	WATCH - cloud ramp risk	POA, GHI, module temp, inverter status, SCADA actual/possible/schedule	cloud_ramp_up on B02/B03/B04
PAV_HUB_4	PAV_S4_B01..B05	250	GOOD	POA, GHI, module temp, inverter status, SCADA actual/possible/schedule	normal historical operation
PAV_HUB_5	PAV_S5_B01..B05	250	GOOD	POA, GHI, module temp, inverter status, SCADA actual/possible/schedule	normal historical operation
PAV_HUB_6	PAV_S6_B01..B05	250	ALERT - curtailment risk	POA, GHI, module temp, inverter status, SCADA actual/possible/schedule	grid_curtailment_plateau on B01-B04
PAV_HUB_7	PAV_S7_B01..B05	250	GOOD	POA, GHI, module temp, inverter status, SCADA actual/possible/schedule	normal historical operation
PAV_HUB_8	PAV_S8_B01..B05	300	GOOD	POA, GHI, module temp, inverter status, SCADA actual/possible/schedule	normal historical operation

4. Data Readiness Matrix

Data area	Available?	Years	Readiness	Notes for agent
SCADA actual/possible/schedule/local limit	Yes	10	High	Can support nowcast, intraday and day-ahead validation.
Block-level inverter status	Yes	8	Medium	Use for equipment derating and block diagnosis.
POA/GHI/module temperature	Yes	5	Medium-high	Sufficient for physical solar model and ramp analysis.

Data area	Available?	Years	Readiness	Notes for agent
Sky-camera cloud motion proxy	Partial	2	Medium	Useful for ramp warning but mark as proxy if exact camera data missing.
NWP ensemble forecast	Yes	3	Medium	Required for day-ahead and weekly uncertainty.
Demand/reserve/penalty data	Partial	2	Medium-low	Needed for business objective and RFP scoring.

5. Event Library for Training and Simulation

Event type	Hierarchy level	Example asset(s)	Business impact	Recommended model response
cloud_ramp_up_after_cloud_clearance	hub/block	PAV_HUB_3, PAV_S3_B02/B03/B04	Fast upward change and timing risk.	Use ramp-rate and cloud obstruction signal; evaluate lead time and phase error.
cloud_ramp_down	hub/block	PAV_HUB_6, PAV_S6_B01/B02/B03	Reserve risk and schedule deviation.	Widen uncertainty band and alert operator before actual drop.
grid_curtailment_plateau	hub/plant/grid	PAV_HUB_6 and plant rollup	Actual clips below possible power.	Classify separately from weather forecast error using local_limit_mw.
scada_stale_signal	block	PAV_S2_B01	Live actual tag cannot be trusted.	Use SCADA quality gate and last-good value; lower confidence.
inverter_group_derating	block	PAV_S4_B05	Equipment-related output loss.	Use availability adjustment and equipment status.

6. Forecast Configuration by Asset Level

Asset level	Use in UI	Forecast aggregation logic	Event detail shown
Grid	Pooling station schedule/deviation view	Sum plant/hub forecasts with correlation-aware uncertainty.	Grid-level schedule and reserve impact.
Plant	Total owner/operator performance	Sum all hubs and compare with plant schedule/local limit.	Plant-level loss and deviation impact.
Hub	Find segment causing issue	Sum blocks inside selected hub plus hub sensors.	Hub rollup and child block events.
Block	Diagnose local issue	Use direct block SCADA and sensors.	Exact block event, inverter and sensor context.

7. Onboarding Agent Expected Output

- A plant readiness summary with missing/available data fields.
- A horizon recommendation table based on business objective and available variables.
- A model selection table by grid/plant/hub/block level.
- A data gap impact section explaining what cannot be enabled confidently.
- RFP requirements for vendors: forecast types, data interfaces, evaluation metrics and uncertainty requirements.
- Clarification questions: dispatch rules, penalty formula, reserve objective, delivery frequency.